Week 9: Housing Data Cleaning & Transformation

(Power BI Desktop)

Submitted by: **Preethi Pal D**

Roll No: **E25DA046**

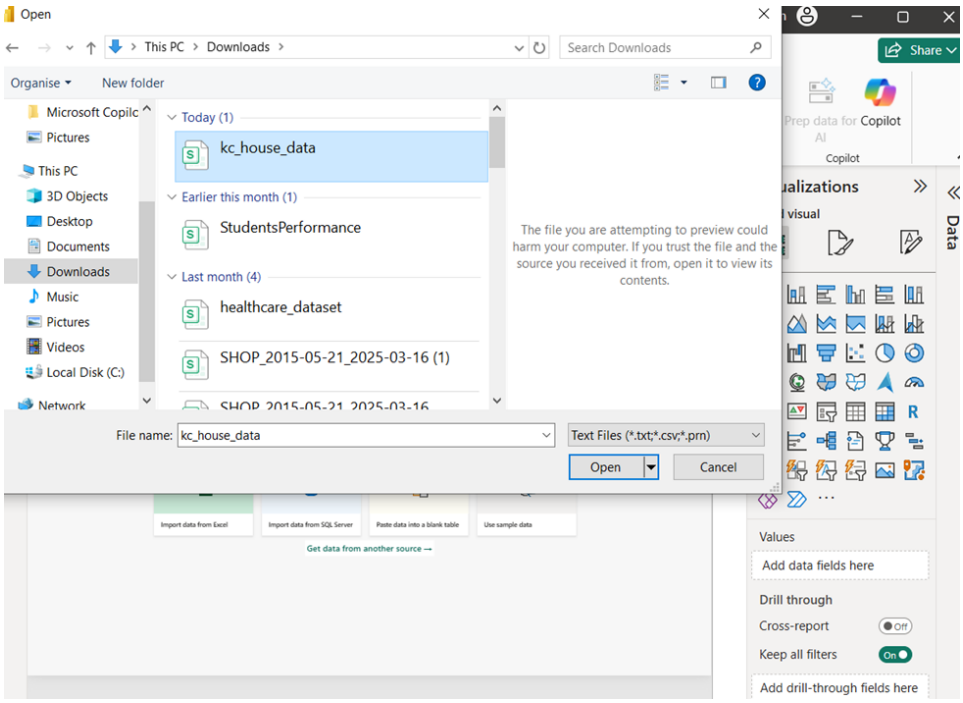
Department: **Bsc Data Science And AI**

Aim

To import the King County house sales data into Power Query, clean it, and create new columns (date parts, property age, price per square foot, size and price categories) so that the data is ready for analysing house prices.

Step 1: Importing the Data

I opened Power BI Desktop and used Get Data with the Text/CSV option to load the file **kc_house_data** from my Downloads folder. After that I clicked Transform Data to open Power Query Editor, where all the cleaning was done.



Step 2: Cleaning the Data

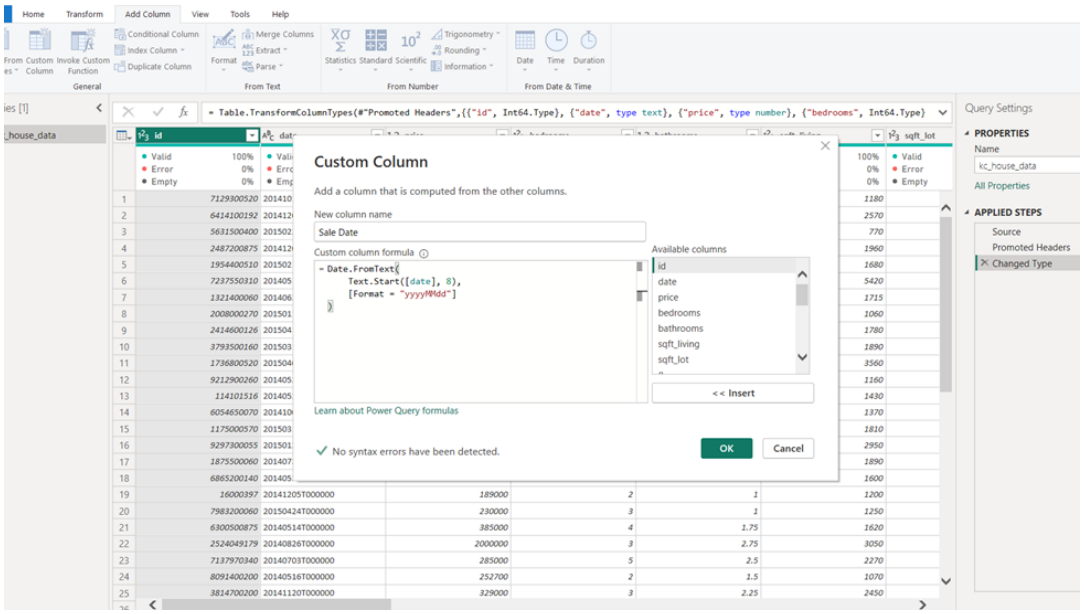
Before making new columns, I cleaned the data in Power Query Editor so that the later calculations are correct. The cleaning steps I did are:

- **Promoted headers** – the first row of the file was used as the column names.
- **Changed data types** – I set each column to the correct type (whole number, decimal number or text) so calculations work properly.
- **Checked column quality** – using the Valid, Error and Empty percentages shown under each column name, I checked for missing or invalid values. The columns showed 100% valid, 0% error and 0% empty.
- **Removed unrealistic rows** – I filtered out the records that have 0 bedrooms or 0 bathrooms, because a house cannot have zero bedrooms or bathrooms.

Step 3: Creating the Sale Date Column

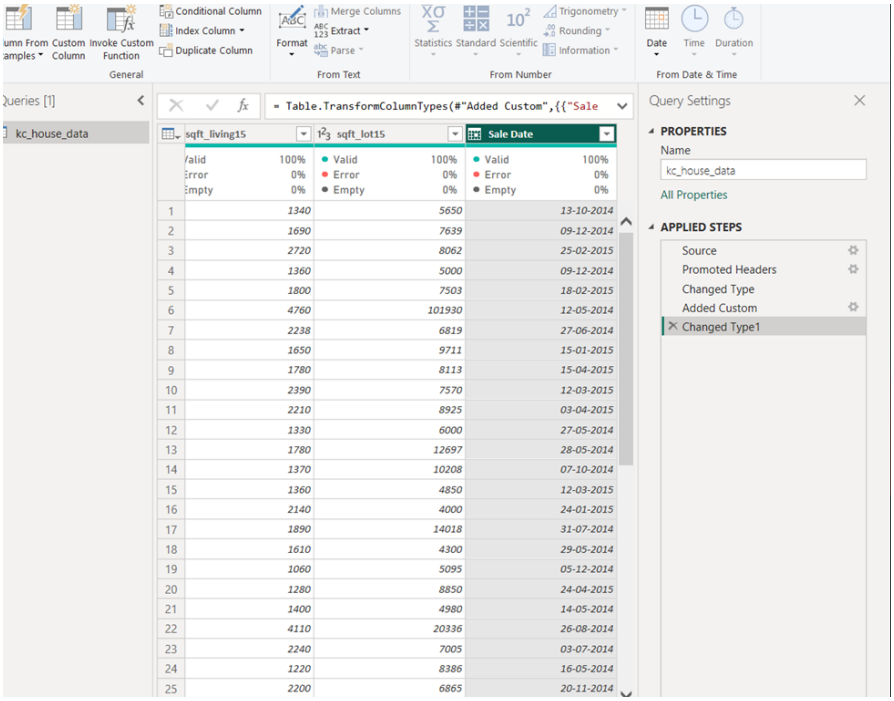
In the dataset the date column is stored as text like 20141013T000000, so it cannot be used as a date directly. I added a custom column named **Sale Date** that takes the first 8 characters of the text and converts them into a date.

- Formula used: Date.FromText(Text.Start([date], 8), [Format = "yyyyMMdd"])



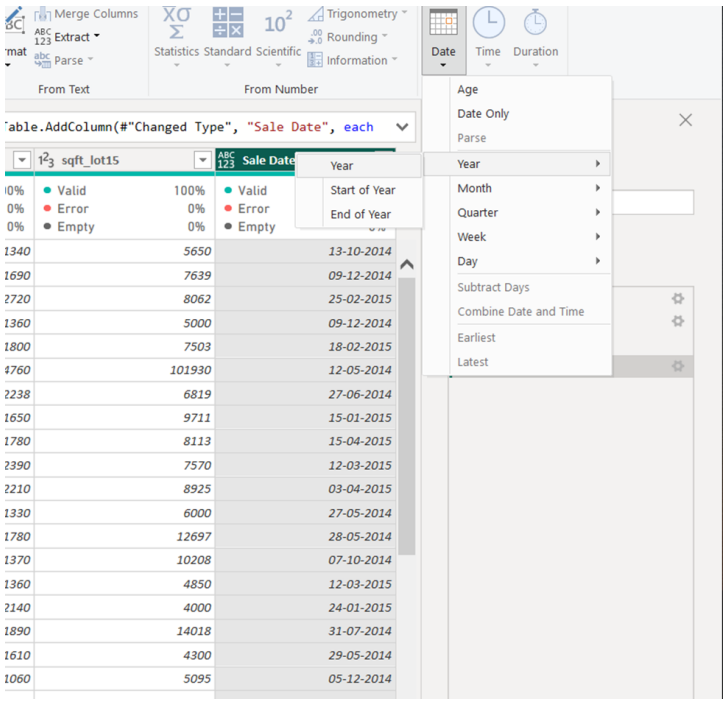
Step 4: Changing the Data Type

Then I changed the data type of Sale Date to Date, so it now shows in a proper date format (for example 13-10-2014). Now the dates can be used for year, month and other date calculations.



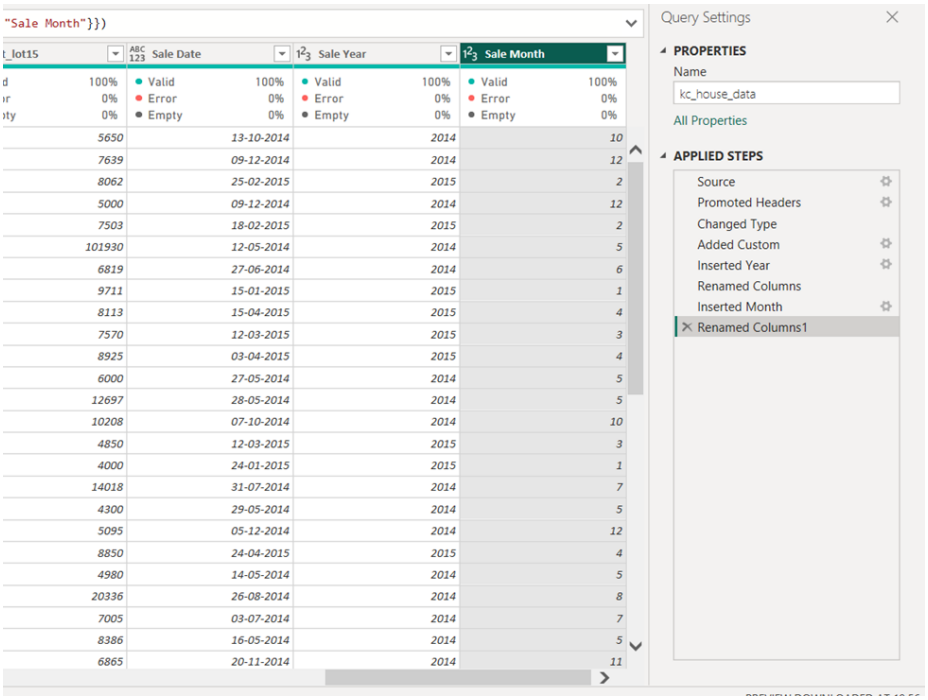
Step 5: Extracting Sale Year and Sale Month

From the Add Column tab I used Date, then Year, to insert the year of the sale. I did the same for Month. I renamed the new columns to **Sale Year** and **Sale Month** so the names are clear.



Step 5 (continued): Sale Year and Sale Month Columns

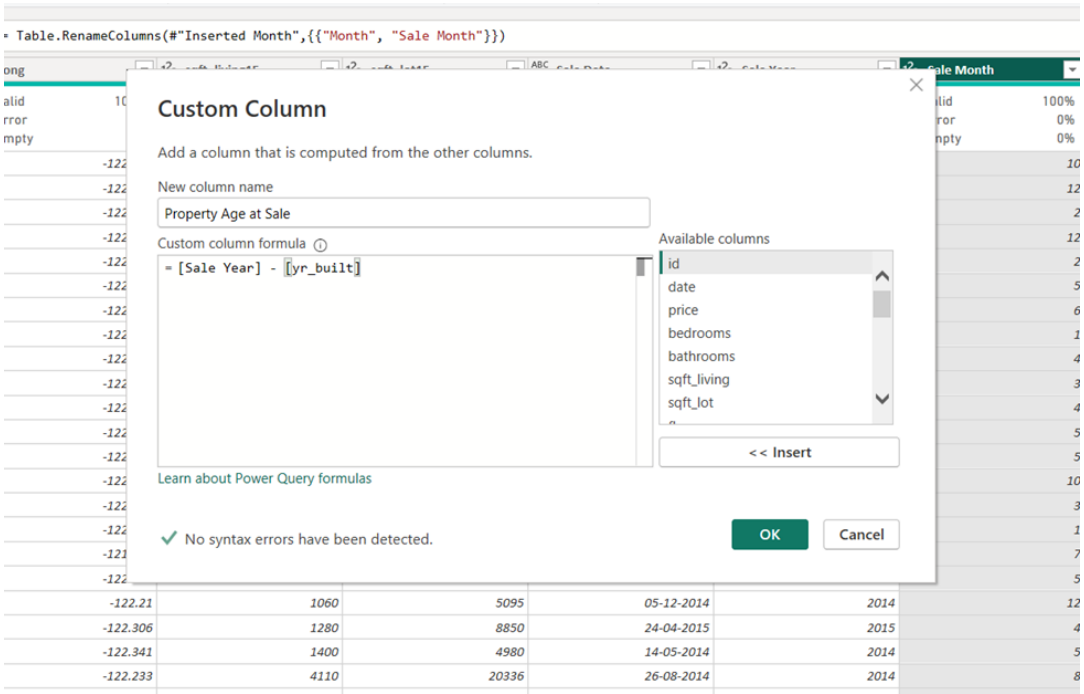
After inserting and renaming, the table has the Sale Year and Sale Month columns next to Sale Date. The Applied Steps pane on the right shows each step I did in order.



Step 6: Property Age at Sale

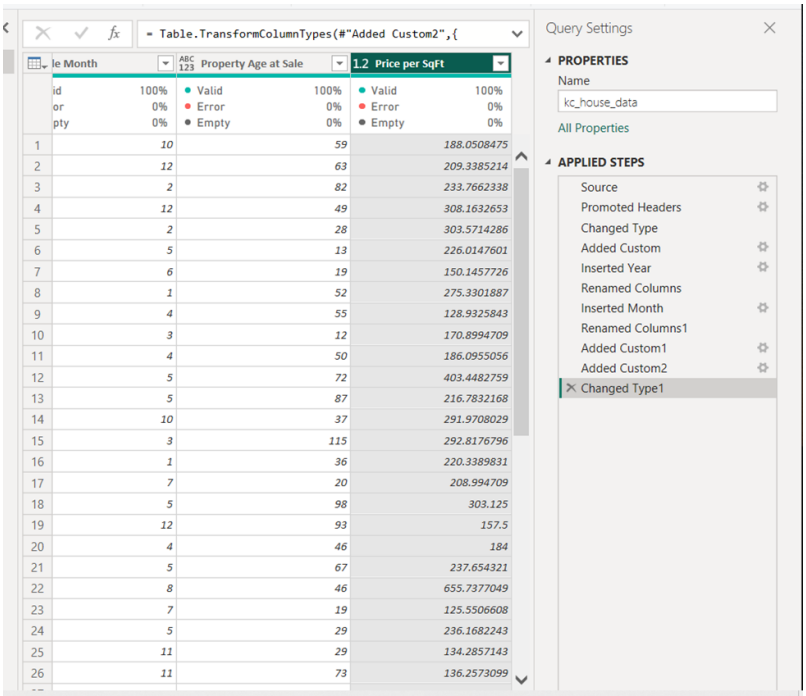
To find how old a house was when it was sold, I added a custom column named **Property Age at Sale**. It subtracts the year the house was built from the year it was sold.

- Formula used: [Sale Year] - [yr_built]



Step 7: Price per SqFt

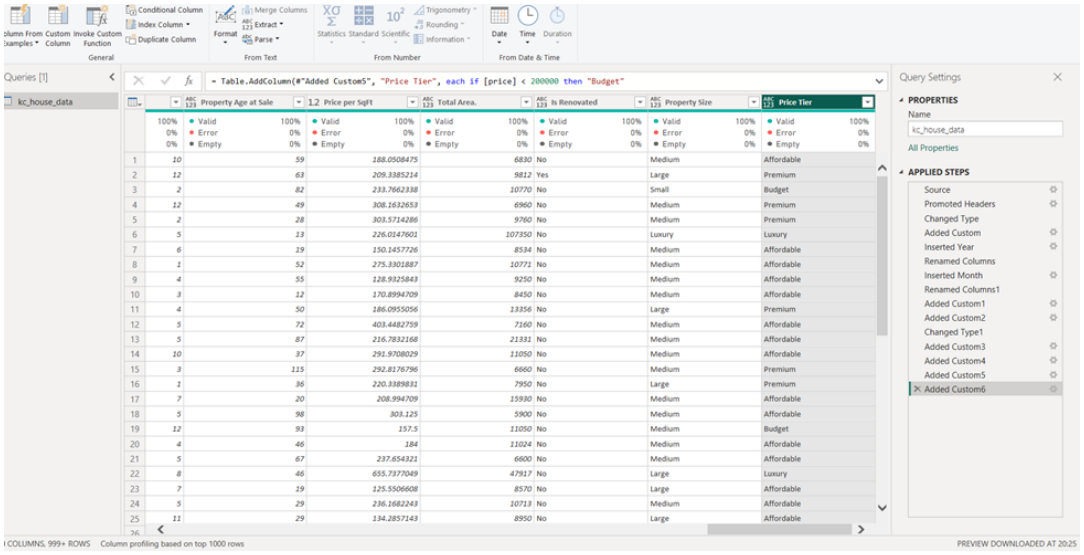
I created a **Price per SqFt** column to compare houses of different sizes fairly. It is the house price divided by the living area, and I set its data type to decimal number.



Step 8: Total Area, Is Renovated, Property Size and Price Tier

I added more custom columns for the analysis:

- **Total Area** – living area plus lot area of the property.
- **Is Renovated** – shows Yes if the house was renovated, otherwise No.
- **Property Size** – groups houses into Small, Medium, Large and Luxury based on their size.
- **Price Tier** – groups houses into Budget, Affordable, Premium and Luxury using an if-then-else condition on the price (for example, a price below 200000 is Budget).



Step 9: Loading the Data

After cleaning and creating all the columns, I clicked Close & Apply to load the cleaned table into the Power BI data model, so the structured housing data is ready to use.

Conclusion

In this week I cleaned and transformed the King County housing data using Power Query. I converted the sale date, extracted the year and month, and created new columns like Property Age at Sale, Price per SqFt, Total Area, Is Renovated, Property Size and Price Tier. The cleaned data is now ready for building reports and finding which factors affect house prices.